

Control and Coordination (Set B) — MCQ Worksheet

Science • Chapter 6 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Nerve impulse along an axon is:

- (A) Electrochemical
- (B) Mechanical only
- (C) Light wave
- (D) Sound wave

Q2. The myelin sheath helps in:

- (A) Speeding up nerve conduction
- (B) Energy production
- (C) Cell division
- (D) Light reception

Q3. Sensory neurons carry signals:

- (A) From receptors to CNS
- (B) From CNS to muscle
- (C) Within muscle
- (D) Outside body

Q4. Motor neurons carry signals:

- (A) From CNS to effector
- (B) From receptor to CNS
- (C) Within sensory organs only
- (D) Inside brain only

Q5. Reflex arc passes through:

- (A) Spinal cord (without going to brain first)
- (B) Cerebrum
- (C) Cerebellum
- (D) Eye only

Q6. Removing hand from a hot object is a:

- (A) Spinal reflex
- (B) Voluntary action
- (C) Hormonal action
- (D) Visual reaction

Q7. The largest part of human brain is:

- (A) Cerebrum
- (B) Cerebellum
- (C) Medulla
- (D) Pons

Q8. Hindbrain controls:

- (A) Heart rate, breathing, blood pressure
- (B) Speech only
- (C) Vision
- (D) Smell only

Q9. Endocrine glands secrete hormones:

- (A) Directly into blood
- (B) Through ducts
- (C) Outside body
- (D) Into nerves

Q10. Thyroid gland secretes:

- (A) Thyroxine
- (B) Insulin
- (C) Adrenaline
- (D) Oestrogen

Q11. Insulin regulates:

- (A) Blood glucose level
- (B) Heart beat
- (C) Sleep
- (D) Vision

Q12. Plant hormone responsible for cell elongation is:

- (A) Auxin
- (B) Cytokinin
- (C) Absciscic acid
- (D) Ethylene

Q13. Hydrotropism is response of plants towards:

- (A) Water
- (B) Light
- (C) Gravity
- (D) Chemicals

Q14. Phototropism in shoots is positive because shoots:

- (A) Grow towards light
- (B) Grow away from light
- (C) Don't respond to light
- (D) Grow down

Q15. Hormone produced in adrenal medulla during stress:

- (A) Adrenaline
- (C) Insulin

- (B) Thyroxine
- (D) Testosterone

Q16. Why are reflex actions faster than voluntary actions?

- (A) They bypass the brain via spinal cord
- (B) Brain processes them faster
- (C) They use eyes only
- (D) Hormones are involved

Q17. Damage to cerebellum causes:

- (A) Loss of balance and coordination
- (B) Loss of memory
- (C) Heart failure
- (D) Vision loss

Q18. Iodine deficiency leads to:

- (A) Goitre due to under-secretion of thyroxine
- (B) Diabetes
- (C) Anaemia
- (D) Rickets

Q19. Why are nervous and hormonal controls said to be complementary?

- (A) Nervous is fast and short-term; hormonal is slower and longer-lasting
- (B) Both same speed
- (C) Both unrelated
- (D) Both are reflexes

Q20. Pollen tube growth towards ovule is an example of:

- (A) Chemotropism
- (B) Phototropism
- (C) Geotropism
- (D) Thigmotropism

Q21. In *Mimosa pudica* (touch-me-not), folding of leaves on touch is due to:

- (A) Sudden loss of water from cells (turgor changes)
- (B) Auxin movement
- (C) Cell division
- (D) Phototropism

Q22. If pituitary is damaged in childhood, the person may develop:

- (A) Dwarfism due to lack of growth hormone
- (B) Diabetes
- (C) Goitre
- (D) Anaemia

Q23. Sensory neuron, interneuron, motor neuron — in order in a reflex arc:

- (A) Receptor → sensory → spinal cord interneuron → motor → effector
- (B) Receptor → motor → sensory → effector
- (C) Effector first
- (D) Brain → cord → effector

Q24. Diabetes is caused by under-secretion of:

- (A) Insulin from pancreas
- (B) Adrenaline
- (C) Thyroxine
- (D) Growth hormone

Q25. Why is hormonal response slow but long lasting?

- (A) Hormones travel via blood to specific targets and act over time
- (B) Hormones never reach cells
- (C) Nerves are faster always
- (D) Hormones don't act